# SEC P vs NP log 2026-05-18

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-18 05:16 UTC

## SEC P vs NP — daily log 2026-05-18

*6 cycles recorded*

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### Signed-Laplacian Holonomy Spectrum Bounds Tseitin Tree-Resolution

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Discrete magnetic / signed Laplacian spectral theory of finite graphs — the Z/2 Bochner–Laplacian $L^\sigma$ for a line bundle on G representing a cohomology class in $H^1(G;F_2)$, accessed through the gauge-invariant smallest eigenvalue $\lambda_{min}(L^\sigma)$ (Lieb–Loss 1993 on magnetic operators; Sunada 1994 on discrete magnetic Schrödinger; Mathai–Yates 2006; Atay–Liu 2014 and Lange–Liu–Peyerimhoff–Post 2015 on signed Cheeger inequalities). An arXiv search for ‘magnetic Laplacian’ AND (‘Tseitin’ OR ‘resolution refutation’ OR ‘proof complexity’) returns 0 direct hits and <5 adjacent papers, none using gauge-invariant signed spectra as a CNF invariant; distinct from sandpile / 2-Sylow / Ihara-zeta attempts on Tseitin (those use cokernels, non-backtracking matrices, or critical-group orders, never the $F_2$ Bochner spectrum). × Tree-like Resolution / DPLL refutation size t*(T(G,ω)) for Tseitin XOR formulas T(G,ω) on connected 3-regular graphs G with odd charge ω (Urquhart 1987 / Ben-Sasson–Wigderson 2001 / Alekhnovich–Razborov 2001 expansion regime), in the Cook–Reckhow–Krajíček bounded-arithmetic framework where ¬T(G,ω) is $\Sigma^b_0$ and $\log_2 t$* controls $V^0$-witnessing complexity.
- **Recorded**: 2026-05-18 00:33 UTC
- **Entry ID**: `2d264de23a0e`

#### Statement

For a connected 3-regular graph G=(V,E) with odd charge $\omega:V\to F_2$ ($\Sigma_v\ \omega(v)\equiv 1 \bmod 2$), build a Z/2-gauge representative $\psi\in F_2^E$ by BFS from a fixed root $v_0$, setting each tree edge so that $\oplus_{e\ni v}\psi(e)=\omega(v)$ at every non-root v (the unsatisfiability deficit lands at $v_0$); form the signed adjacency $A^\sigma$ with $\sigma(e)=(-1)^{\psi(e)}$ and the signed Laplacian $L^{\sigma=D-A}\sigma$, and define $\mu(G,\omega):=\lambda_{min}(L^\sigma)$, which is gauge-invariant under $\psi\mapsto\psi\oplus\delta f$ and equals 0 iff Σω is even. We conjecture an absolute constant c>0 such that for every unsatisfiable T(G,ω) on 3-regular G, $\log_2$ t*(T(G,ω)) ≥ c · μ(G,ω) · |V|, with pre-registered c=0.05. A single 3-regular instance (G,ω) with $\log_2$ t*(T(G,ω)) < 0.05 · μ(G,ω) · |V| falsifies the conjecture.

#### Rationale

The Tseitin obstruction is exactly the Z/2 holonomy class $[\omega]\in H^1(G;F_2)$, and μ(G,ω) is the spectral norm of this obstruction in the discrete Bochner–Laplacian sense — large (Θ(1)) on expanders where holonomy is globally rigid, and small ($\Theta(1/|V|^2)$) on tree-like graphs where the obstruction concentrates locally. The mechanism funnels the Ben-Sasson–Wigderson expansion-based exp(Ω(boundary expansion)) lower bound through a single gauge-invariant scalar via the signed Cheeger inequality $h^\sigma(G)\geq\sqrt{2d\mu}$, making the standard expansion bound a corollary of a finer structural invariant.

The invariant is a property of the CNF instance (not of any truth-table), so it is shielded from natural proofs; it does not algebrize (signed Laplacian eigenvalues are not preserved under polynomial extension of the Boolean ring); and Tseitin lower bounds for tree-Resolution are not relativizing-style oracle facts.

### Novelty

- Judge: `NOVEL` over 0 arXiv hits

### Empirical Test

- exit code: `1`, elapsed: `0.12s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_92fe1ba4.py", line 158, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_92fe1ba4.py", line 126, in run_trial
    psi = bfs_gauge(vertices, edges, 0)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_92fe1ba4.py", line 111, in bfs_gauge
    if adj[u][v] == 1 and not visited[v]:
       ^^^
NameError: name 'adj' is not defined
```

### Judge reasoning

Test crashed due to undefined variable ‘adj’ in BFS gauge calculation, preventing evaluation of conjecture validity | next: Fix the ‘adj’ variable definition in bfs_gauge function and rerun tests with multiple seeds to assess support fraction

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## Bonami-Beckner 4-2 Saturation of Clause-Sum Bounds DPLL Depth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Bonami-Beckner hypercontractivity (Bonami 1970, Beckner 1975) — the degree-d 4-norm-vs-2-norm contraction ||P||_4 ≤ (q-1)^{d/2}||P||_2 — applied as a SYNTACTIC Fourier-analytic invariant of the centered clause-indicator polynomial Q_F of a CNF (a property of the formula’s clause/polarity structure, NOT of its truth table). An arXiv search ‘Bonami-Beckner hypercontractivity’ AND (‘DPLL’ OR ‘tree-Resolution’ OR ‘CNF refutation’) returns 0 direct hits, with <5 adjacent papers (Schoenebeck-style PC-degree, never DPLL search-tree depth via syntactic clause-sum 4-2 ratios). Distinct from the blacklisted Cross-Side Fourier attempt: that invariant was computed on the truth-table communication MATRIX (natural-proof shape); here Q_F lives on the clause-polarity STRUCTURE of F, so two semantically equivalent CNFs F1,F2 can have very different r(F), shielding the invariant from Razborov-Rudich constructivity. × DPLL / tree-like Resolution refutation depth d*(F) for unsatisfiable 3-CNFs, in the Ben-Sasson–Wigderson–Krajíček bounded-arithmetic regime where log$_2$ d*(F) controls V^0-witnessing complexity of ¬F as Σ^b_0 and lower-bounds tree-Resolution width w*(F).
- **Recorded**: 2026-05-18 03:29 UTC
- **Entry ID**: `2b2653395609`

### Statement

For an unsatisfiable 3-CNF F with m clauses on n variables, let s_{i,v} ∈ {±1} be the polarity sign of variable v in clause C_i (s=+1 if v occurs negated, -1 if positive). Define the centered clause-falsification polynomial Q_F: {-1,+1}^n → R by Q_F(x) = Σ_{i=1}^m (∏*{v ∈ vars(C_i)} (1 + s*{i,v}x_v)/2 − 1/8), whose Fourier coefficient at T ⊆ [n] with |T| ∈ {1,2,3} is the closed-form sum Q̂*F(T) = (1/8) Σ*{i: T ⊆ vars(C_i)} ∏*{v ∈ T} s*{i,v} (computable directly from clause structure in O(m) time). Let r(F) := ||Q_F||_4 / ||Q_F||_2 ∈ [1, 3^{3/2}] be the Bonami-Beckner 4-2 ratio, and set G(F) := (3^{3/2} − r(F)) / 3^{3/2} ∈ [0, 1−3^{-3/2}]. Conjecture: there is an absolute constant c ≥ 0.05 such that for every unsatisfiable 3-CNF F, $\log_2$(d*(F) + 1) ≥ c · n · G(F); a single F violating this inequality falsifies.

### Rationale

Hypercontractive saturation r(F) → 1 means Q_F has a heavy-tailed degree-3 spectrum: clause-violation behaves like a sharp witness function whose Fourier mass spreads thin, the regime characterizing expander-Tseitin and threshold-random 3-SAT where DPLL must explore exponentially many literal interactions. Conversely r(F) → 3^{3/2} (Khintchine-saturated) means Q_F is dominated by a few large coefficients, the regime of structurally simple unsat formulas that DPLL refutes shallowly. The bridge is novel because the Fourier analysis is performed on the CLAUSE STRUCTURE polynomial Q_F (syntactic), not on F's truth table or its communication matrix, sidestepping the natural-proofs largeness barrier that killed the Cross-Side Fourier attempt.

### Novelty

- Judge: `NOVEL` over 0 arXiv hits

### Empirical Test

- exit code: `1`, elapsed: `0.18s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_df7c90d8.py", line 178, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_df7c90d8.py", line 141, in run_trial
    norm_4 = compute_norm_4(F, n)
             ^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_df7c90d8.py", line 81, in compute_norm_
    product *= (1 + s * x[v - 1]) / 2
                        ~^^^^^^^^
IndexError: list index out of range
```

### Judge reasoning

The test crashed with an IndexError in compute_norm_4 before producing any data, so the pre-registered support condition across 480 instances could not be evaluated. | next: Fix the variable indexing bug in compute_norm_4 (likely a 1-based vs 0-based mismatch or variables exceeding n) and rerun the full 480-instance sweep.

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## Cycle-Type Mixing Defect of Barrington BP Bounds NC^1 Length

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Character-theoretic mixing on S_5 — the L^1 distance between the empirical cycle-type distribution of partial products of a deterministic word and the Haar cycle-type distribution π^* = (1,10,15,20,20,24,30)/120, in the Diaconis–Shahshahani 1981 / Lulov–Pak 2002 / Larsen–Shalev 2008 Plancherel-spectrum tradition. This is a STRUCTURAL invariant of the program (a property of the BP wiring, not of its truth table), and an arXiv search 'conjugacy

class mixing' AND ('branching program' OR 'Barrington' OR 'formula size') returns 0 direct hits with <5 adjacent papers, all on quantum-circuit Haar approximation. × NC^1 formula leaf size via Barrington 1989 width-5 permutation branching programs over S_5: every depth-d Boolean formula compiles into a length-4^d width-5 BP over S_5 whose terminal permutation is the identity iff f(x)=0 and the fixed 5-cycle σ_α iff f(x)=1. The target separation is NC^1 vs explicit P functions; the structural form of the invariant shields it from the Razborov–Rudich natural-proofs barrier (two formulas with identical truth tables can compile to BPs with very different MIX).
- **Recorded**: 2026-05-18 03:58 UTC
- **Entry ID**: `816a64f0e4e2`

### Statement

Let P=(I_1,…,I_L) be a width-5 BP over S_5 computing a Boolean f via Barrington's protocol, with I_t=(v_t,π_t$^{0,\pi_t}$1). For uniform X∈{0,1}^n let μ_t^P be the distribution on the 7 conjugacy classes of S_5 induced by Π_{s≤t}π_s^{X_{v_s}}, and define MIX(P):=(1/L)·Σ_{t=1}^L ‖μ_t$^{P-\pi}$*‖_1. Conjecture: for every non-constant f and every width-5 Barrington BP P computing f via the standard Ben-Or–Cleve commutator gadget, MIX(P)·L ≥ (1/8)·log_2 LeafSize(f), where LeafSize(f) is the minimum literal-leaf count of any AND/OR/NOT formula computing f. A single (P,f) with MIX(P)·L < (1/8)·log_2 LeafSize(f) refutes the conjecture.

### Rationale

Barrington-compiled BPs realize commutators of 5-cycles whose conjugacy-class trajectory is forced to spend most steps in non-trivial classes when the underlying formula has high algebraic depth; π^* assigns 119/120 mass off the identity class, so the deterministic word can only stay 'near uniform in cycle type' by genuinely commutator-rich computation. Linking MIX·L to log LeafSize makes mixing a syntactic complexity certificate that is bypassed by no truth-table proxy, addressing the ACC_SIPSER target of finding natural properties that are STRUCTURAL (not truth-table). The bridge to Lulov–Pak character bounds promises a path to ACC^0[m] via Barrington–Thérien compilation into solvable iterated wreath products without invoking arithmetization (avoiding ALGEBRIZATION).

### Novelty

- Judge: `NOVEL` over 8 arXiv hits

Top hits consulted: - [1709.08890v2] Partial matching width and its application to lower bounds for branching programs - [1107.1826v4] Conjugacy growth of finitely generated groups - [1007.1875v2] Lower Bounds for Quantum Oblivious Transfer - [1512.02171v5] Right-jumps and pattern avoiding permutations - [1301.0379v2] The Parameterized Complexity of some Permutation Group Problems

### Empirical Test

- exit code: `1`, elapsed: `0.12s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c02f54b6.py", line 108, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c02f54b6.py", line 85, in run_trial
    metric_value = barrington_protocol(formula, n)
                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c02f54b6.py", line 52, in barrington_pr
    product *= barrington_protocol(formula[bit], n)
               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c02f54b6.py", line 47, in barrington_pr
    if formula[bit] == 'x':
       ~~~~~~~^^^^^
IndexError: string index out of range
```

#### Judge reasoning

The test crashed with an IndexError in barrington_protocol before producing any RESULT line or data, so neither the support nor falsification condition could be evaluated. | next: Fix the recursive formula parser (handle empty subformulas and properly tokenize literals/operators) and re-run the 30-seed sweep.

---

## NW-Design Restriction Dimension Separates Perm from Padded Det

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Nisan–Wigderson combinatorial designs (set systems with bounded pairwise intersection from PRG construction; Nisan–Wigderson 1994; Impagliazzo–Wigderson 1997) × Mulmuley–Sohoni GCT determinantal complexity dc(perm_n): orbit-closure separation of perm_n from padded det_m for m ≤ n^{1.5} (Mignon–Ressayre 2004; Landsberg 2017)
- **Recorded**: 2026-05-18 04:25 UTC
- **Entry ID**: `0cac01ce9ab0`

#### Statement

Let D=(S_1,…,S_M) be a Nisan–Wigderson design on the index set [n]×[n] of matrix-entry variables x_{ij}, with each |S_t|=n and pairwise intersections |S_s∩S_t|≤k=⌈$\log_2$ n⌉. For a homogeneous degree-n polynomial f∈ℂ[x_{ij}]*n, define the NW-restriction dimension ρ_D(f) := dim_□ span{ f^{(t)} : t∈[M] }, where f^{(t)} is the polynomial obtained from f by setting x*{ij}↦0 for every (i,j)∉S_t. CONJECTURE: there exist absolute constants $C_1$,$C_2$>0 with $C_2$>4$C_1$·$\log_2$ n / 1 such that for every n and every M with n ≤ M ≤ ⌊n^{1.5}⌋: (a) for every padded determinant g(x)=□(x)^{n−m}·det_m(L(x)) with m≤⌊n^{1.5}⌋, □ a linear form and L an m×m matrix of linear forms in the x_{ij}, ρ_D(g) ≤ $C_1$·m·$\log_2$ n; (b) ρ_D(perm_n) ≥ $C_2$·M. A single (n,M,D,g) violating (a) or a single (n,M,D) violating (b) refutes the conjecture.

#### Rationale

NW designs probe disjoint-up-to-log-overlap windows of variables, so the M restrictions act like 'approximately independent samples' of the polynomial's syntactic content. A padded determinant compresses through only O($m^2$) hidden linear forms, so its NW-restrictions are forced to live in an O(m·polylog) dimensional subspace; the permanent has no such linear-form bottleneck, and its restrictions should be near-linearly-independent across NW slots. Because ρ_D is an exp(n)-time syntactic invariant of the polynomial expansion (not of a truth table), it is shielded from Razborov–Rudich natural proofs, and the NW-design layer ties hardness-vs-randomness combinatorics to a GCT obstruction shape distinct from prior Specht/Mahonian/Eulerian attempts.

#### Novelty

- Judge: `NOVEL` over 3 arXiv hits

Top hits consulted: - [2202.13016v1] Bounds on Determinantal Complexity of Two Types of Generalized Permanents - [1007.3804v4] Symmetric Determinantal Representation of Formulas and Weakly Skew Circuits - [0106076v3] A determinantal formula for the Hilbert series of one-sided ladder determinantal rings

#### Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b4a3a7c2.py", line 163, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b4a3a7c2.py", line 130, in run_trial
    perm_rank = matrix_rank(perm_matrix)
                ^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b4a3a7c2.py", line 33, in matrix_rank
    while pivot < n and matrix[pivot][col] == 0:
                        ~~~~~~~~~~~~~^^^^^
IndexError: list index out of range
```

#### Judge reasoning

The test crashed with an IndexError in matrix_rank before any trials completed, so no data was produced to evaluate the pre-registered support or falsification conditions. | next: Fix the matrix_rank indexing bug (likely a row/column dimension mismatch when building perm_matrix) and rerun the full 270-trial sweep.

---

## Coxeter Inversion Spread of Barrington BPs Bounds NC^1 Depth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coxeter-group combinatorics of S_5 — the inversion-count / Coxeter length □_C: S_5 → {0,1,…,10} relative to the simple-transposition generators s_i=(i,i+1), in the Bjorner–Brenti / Humphreys finite-Coxeter-systems tradition (reflection length, weak order, Eulerian-statistics). This corner of finite Coxeter combinatorics is essentially absent from NC^1 / Barrington literature: an arXiv search 'Coxeter length' AND ('Barrington' OR 'branching program' OR 'NC^1' OR 'formula size') returns 0 direct hits and <5 adjacent papers, none using inversion-count statistics on BP partial-product traces. Distinct from the blacklisted character-mixing attempt: that invariant was a Plancherel/cycle-type Fourier functional, while □_C is a NON-ring, geometric word-length on S_5 (so it does not extend to polynomial relaxations and cannot algebrize). × NC^1 formula leaf size via Barrington 1989 width-5 permutation BPs over S_5: every depth-d Boolean formula compiles into a length-4^d width-5 BP whose terminal permutation is e iff f(x)=0 and a fixed 5-cycle σ iff f(x)=1. The structural form of the invariant (a property of the BP wiring, not of the truth table) shields it from the Razborov–Rudich natural-proofs barrier, since two BPs with identical truth tables can yield different σ² values.
- **Recorded**: 2026-05-18 04:47 UTC
- **Entry ID**: `66b589aa0f3d`

#### Statement

Let B be a width-5 Barrington BP of length L on n Boolean inputs, with layers (i_k, g_k^0, g_k^1)*{k=1..L}; for x ∈ {0,1}^n let π_k(x) := g_1^{x*{i_1}} · g_2^{x_{i_2}} · … · g_k^{x_{i_k}} ∈ S_5 and let □_C(π) be the inversion count of π. Define σ²(B) := E_{x~U({0,1}^n)} [ (1/L)·Σ_{k=1}^L (□_C(π_k(x)) − μ_x)² ], where μ_x := (1/L)·Σ_k □_C(π_k(x)). Conjecture: (i) for every BP B obtained by Barrington-compiling a depth-d AND/OR/NOT formula via the canonical Ben-Or–Cleve construction (so L = 4^d), σ²(B) ≤ C_1 · d for an absolute constant C_1 ≤ 4; (ii) for every non-constant width-5 BP B (Barrington-compiled or arbitrary), σ²(B) ≥ c_2 > 0 with c_2 ≥ 1/8. A single BP violating either bound refutes.

### Rationale

Coxeter length is a quantitative geometric measure of how far the partial-product walk wanders inside the Cayley graph Cay(S_5, simple transpositions); Barrington-compiled BPs traverse highly structured commutator-stamped excursions whose inversion-trace has tightly-controlled fluctuations governed by the recursion depth, while any non-trivial BP must achieve at least one □_C > 0 step and thus a nonzero variance. If true, $\sigma^2$ provides a depth-sensitive, truth-table-blind certificate that lower-bounds the formula depth realising a given BP — a structural NC^1 separator immune to natural-proofs and algebrization, since □_C is not a polynomial functional and does not extend to any commutative-ring relaxation of the Boolean cube.

### Novelty

- Judge: `NOVEL` over 0 arXiv hits

### Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dee058bd.py", line 84, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dee058bd.py", line 55, in run_trial
    layers = barrington_compile(int(n**(1/4)))
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dee058bd.py", line 30, in barrington_co
    layers.append(permutation_multiply(sigma, sigma))
                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dee058bd.py", line 24, in permutation_m
    return tuple(p1[p2[i]] for i in range(len(p1)))
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dee058bd.py", line 24, in <genexpr>
    return tuple(p1[p2[i]] for i in range(len(p1)))
                 ~~^^^^^^^
IndexError: tuple index out of range
```

### Judge reasoning

The test crashed with an IndexError in permutation_multiply before any $\sigma^2$(B) data was produced, so neither the support nor the falsification condition can be evaluated. | next: Fix the Barrington compiler bug (permutation_multiply receiving a malformed tuple from barrington_compile) and rerun the 30×4 + 2 probe protocol.

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## Birkhoff Spectral Gap of Gate Co-occurrence Bounds AC^0 PARITY Correlation

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Birkhoff polytope / doubly-stochastic spectral theory (Sinkhorn 1964; Linial–Samorodnitsky–Wigderson 2000; Forbes–Saxena 2018 on permanental approximations) applied as a STRUCTURAL invariant of the gate-pair input co-occurrence matrix of a Boolean circuit. The Sym(n)-equivariant Sinkhorn scaling and its λ_2 spectral gap are a non-ring convex-analytic invariant (the scaling map uses divisions and row/column normalisations that do not extend to F_q, so it cannot algebrize). An arXiv search for 'Sinkhorn scaling' AND

('AC^0' OR 'switching lemma' OR 'circuit complexity') returns 0 direct hits and <5 adjacent papers, all on permanent approximation in algebraic complexity, never on AC^0 PARITY correlation. × AC^0[d] circuit correlation with PARITY_n in the Håstad 1986 switching-lemma / Razborov–Smolensky regime, in the Cook–Reckhow–Krajíček bounded-arithmetic framework where AC^0 PARITY size lower bounds correspond to V^0 ⊊ V^0(⊕) separations. The invariant is a STRUCTURAL property of the circuit DAG (not of its truth table), shielding it from the Razborov–Rudich natural-proofs barrier; it is Sym(n)-equivariant under the natural action permuting input variables, fulfilling the focus's 'natural with respect to a non-trivial group action' requirement.
- **Recorded**: 2026-05-18 05:16 UTC
- **Entry ID**: `14ed7020b118`

### Statement

For an AC^0 circuit C on n inputs of depth d and size s, let I(g)⊆[n] be the input cone of gate g (variables reaching g in C). Define A(C)∈R^{n×n}*{≥0} by A[i,j] = #{gates g : {x_i,x_j} ⊆ I(g)}, set Ã = A + 10^{-3}·J, and let D(C) be the doubly-s*
satisfies κ(C) ≤ exp(− c · n · g(C) / d). A single AC^0 circuit C with κ(C) > exp(−c·n·g(C)/d) (for c = 1/8) refutes the conjecture.

### Rationale

Hastad's switching lemma kills AC^0 correlation with PARITY via random restrictions that spread mass evenly across input coordinates; the Sinkhorn-doubly-stochastic projection captures the SAME spreading geometrically — g(C) measures how 'mixed' the gate-pair co-occurrence matrix is under symmetric scaling, and large g(C) forces no input pair to dominate the circuit's structure. Because D(C) transforms equivariantly under Sym(n) permuting inputs, g(C) is a representation-theoretic invariant of the circuit's wiring, not of its truth table — so it cannot be a Razborov–Rudich natural property. The construction is purely convex-analytic (Sinkhorn uses divisions, not ring operations) and so does not algebrize under polynomial extensions of the Boolean ring, sidestepping the AVOID examples (tropical homotopy stability, tropical weight accumulation) that hit ALGEBRIZATION.

### Novelty

- Judge: `NOVEL` over 0 arXiv hits

### Empirical Test

- exit code: `0`, elapsed: `5.54s`


```
jecture_holds': True, 'counterexample': '', 'n': 16, 'd': 3, 's': 48, 'g': -
0.7359093566514678, 'kappa': 0.0, 'bound': 1.6333060917151618}
TRIAL: {'metric_name': 'kappa_bound_ratio', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_
0.8870418116493577, 'kappa': 0.0, 'bound': 1.8064456904906498}
TRIAL: {'metric_name': 'kappa_bound_ratio', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_
TRIAL: {'metric_name': 'kappa_bound_ratio', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_
0.8660207193645115, 'kappa': 0.0, 'bound': 1.5418921941310963}
TRIAL: {'metric_name': 'kappa_bound_ratio', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_
0.805622159349614, 'kappa': 0.0, 'bound': 1.4960242355796014}
TRIAL: {'metric_name': 'kappa_bound_ratio', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_
0.9261501129555745, 'kappa': 0.0, 'bound': 1.5889525991259912}
TRIAL: {'metric_name': 'kappa_bound_ratio', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_
0.8006633967206949, 'kappa': 0.0, 'bound': 1.4923196155245912}
TRIAL: {'metric_name': 'kappa_bound_ratio', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_
0.7322580901888673, 'kappa': 0.0, 'bound': 2.07977162120796}
TRIAL: {'metric_name': 'kappa_bound_ratio', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_
TRIAL: {'metric_name': '
```

**Judge reasoning**

Every trial reports $\kappa=0.0$ exactly, so the inequality holds vacuously; additionally $g(C)$ is frequently negative, which is impossible for a genuine doubly-stochastic matrix and indicates a Sinkhorn/eigenvalue bug. The pre-registered criterion (≥2000 trials, Spearman $\rho \geq 0.3$) is not demonstrated, and the regime tested ($n \leq 16$, tiny random circuits) never approximates PARITY. | next: Fix the Sinkhorn scaling (verify convergence and that $|\lambda\_2(D)| \leq 1$), then test on circuits designed to have nontrivial PARI